CIE AS PHYSICS (9702) • QUESTION PAPER

1.4 Scalars & Vectors

Syllabus Section: 1. Physical Quantities & Units **Paper Type:** Multiple Choice (Paper 1) **Total:** 16 Questions (16 Marks)

Question 1 Easy [1 mark]

Which pair contains one vector and one scalar quantity?

A	Displacement	Acceleration
B	Force	Kinetic energy
C	Momentum	Velocity
D	Power	Speed

A

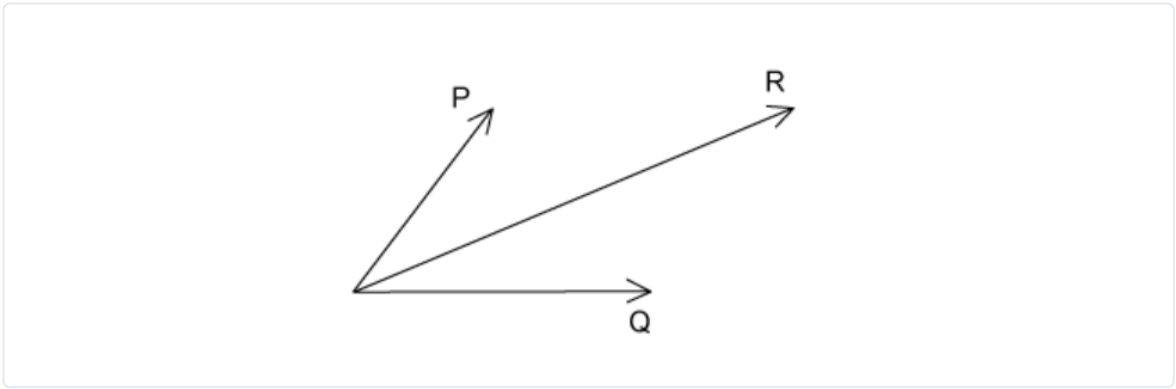
B

C

D

Question 2 Easy [1 mark]

Two physical quantities **P** and **Q** are added together. The sum of **P** and **Q** is **R**, as shown in the diagram.



Which quantity could be represented by **P** and by **Q**?

A Kinetic energy

B Power

C Speed

D Velocity

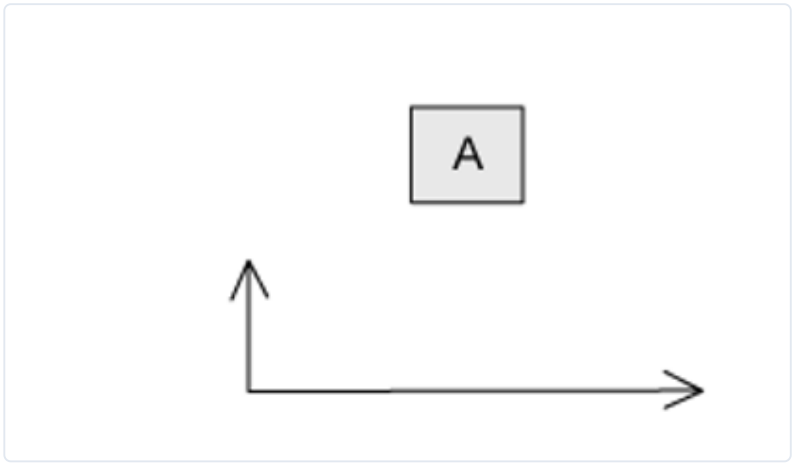
Question 3 Easy [1 mark]

The arrow represents the vector **R**.

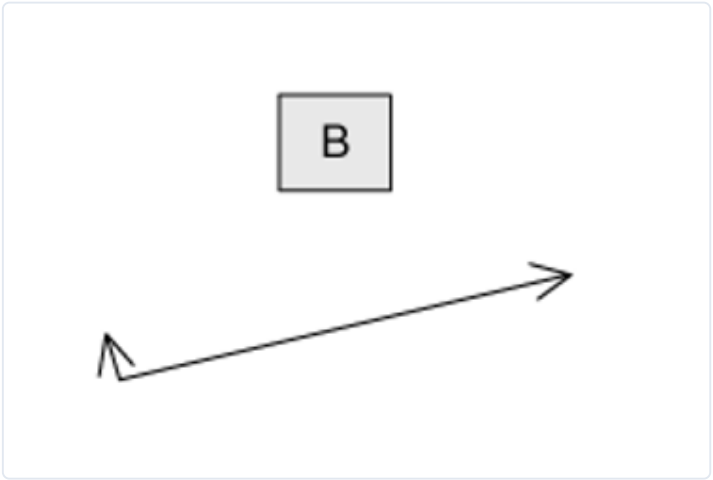


Which diagram does **not** represent **R** as two perpendicular components?

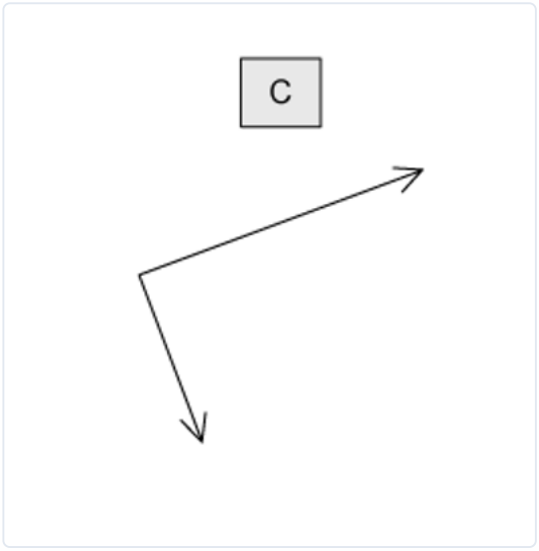
A



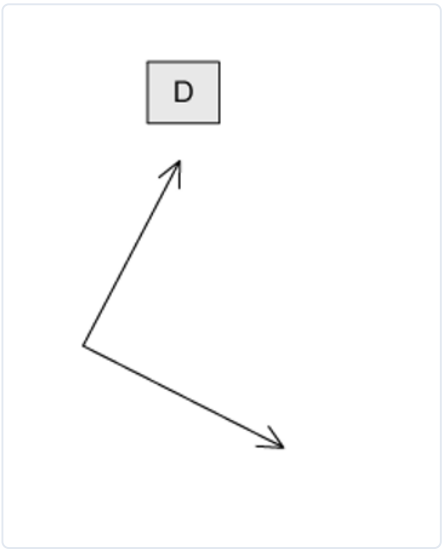
B



C

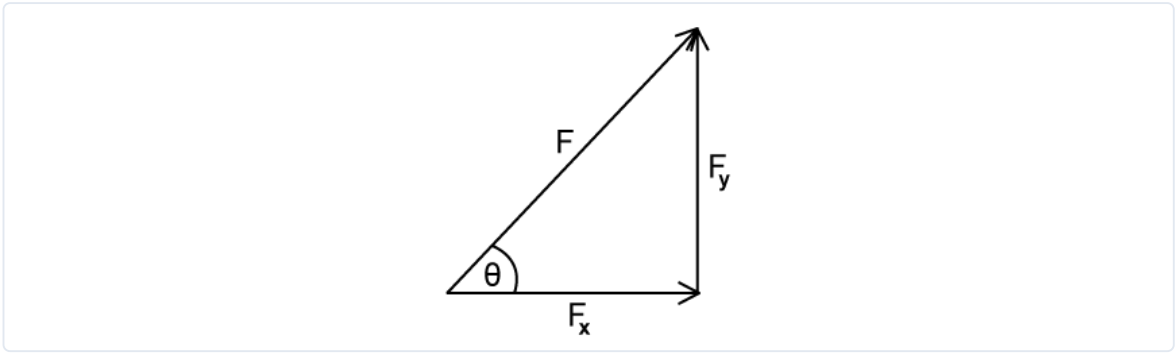


D



Question 4 Easy [1 mark]

The diagram shows a resultant force *F* at an angle of *θ* to the horizontal. The horizontal component F_x and vertical component F_y are labelled on the diagram.



What is the horizontal component of *F*?

A $F \sin \theta$

B $F \cos \theta$

C $F \tan \theta$

D $F_y \cos \theta$

Question 5 Easy [1 mark]

Which one of the following statements is **true**?

A A scalar quantity has both magnitude and direction

B A scalar quantity has only direction

C A vector quantity has only magnitude

D A vector quantity has both magnitude and direction

Question 6 Medium [1 mark]

The speed of an aeroplane in still air is 200 km h^{-1}. The wind blows from the west at a speed of 85.0 km h^{-1}.

In which direction must the pilot steer the aeroplane in order to fly due north?

A 23.0° east of north

B 23.0° west of north

C 25.2° east of north

D 25.2° west of north

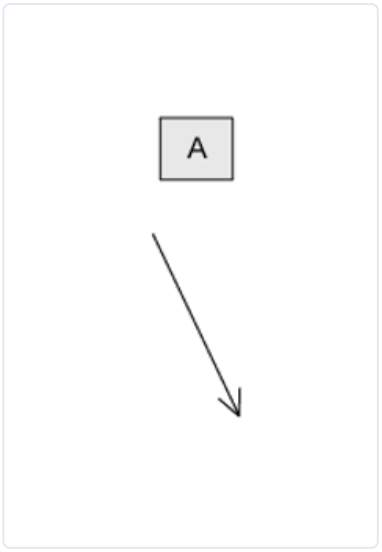
Question 7 Medium [1 mark]

Vectors **P** and **Q** are drawn to scale.

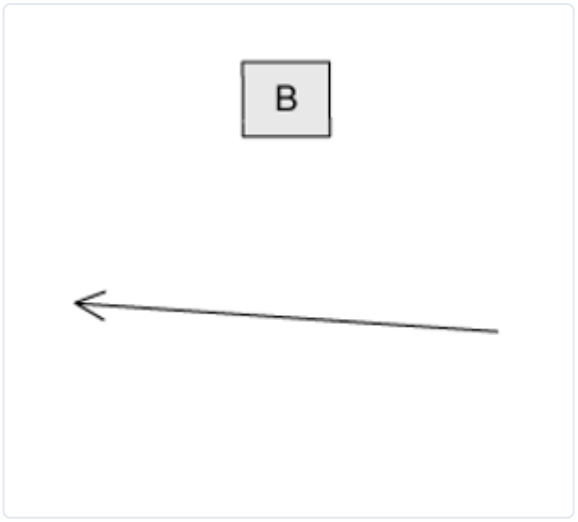


Which diagram represents the vector (**P** – **Q**)?

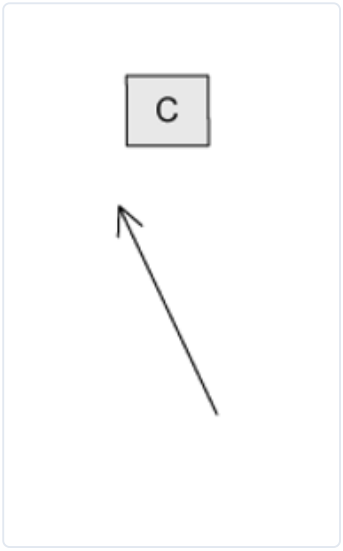
A



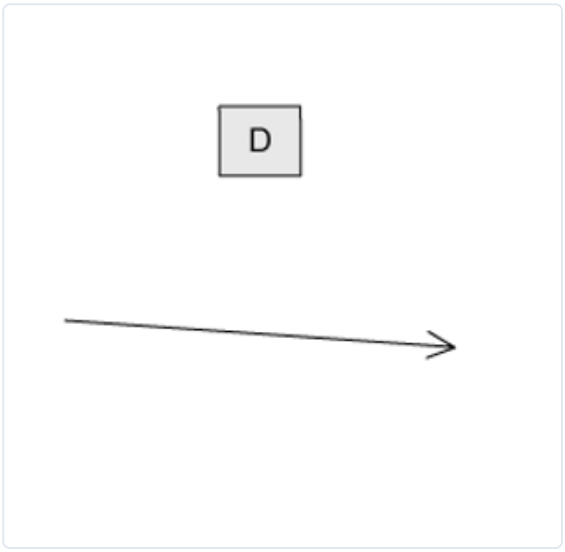
B



C

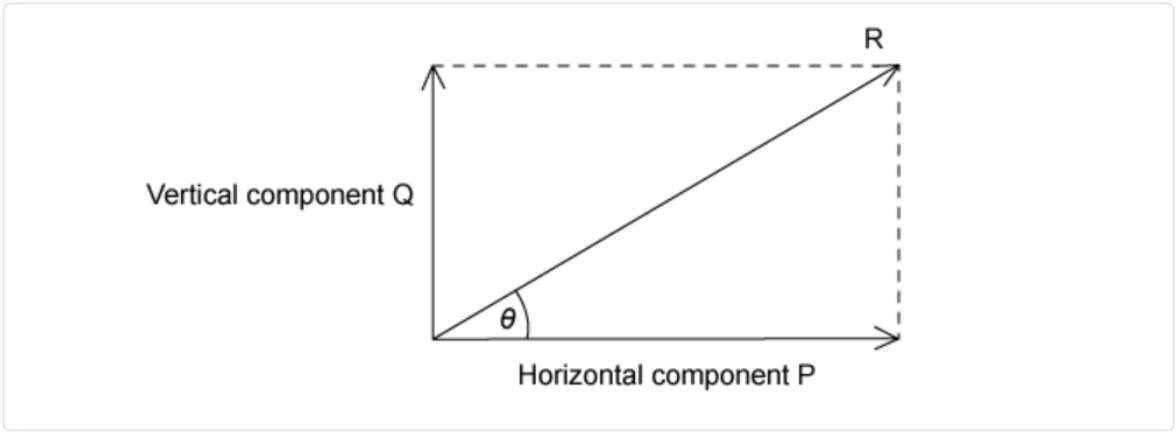


D



Question 8 Medium [1 mark]

A vector has magnitude **R** and perpendicular components **P** and **Q**, as shown in the diagram.



Which row correctly describes the perpendicular components?

	Vertical component	Horizontal component
A	Q	$\sin\theta$
B	$R\cos\theta$	P
C	$R\cos\theta$	$R\sin\theta$
D	$R\sin\theta$	$R\cos\theta$

A

B

C

D

Question 9 Medium [1 mark]

A resultant force of 20 N is 30° to the horizontal.

What is the horizontal and vertical component of the resultant force?

	Horizontal Component / N	**Vertical Component / N**
A	17	10
B	10	17
C	11.5	10
D	17	11.5

A

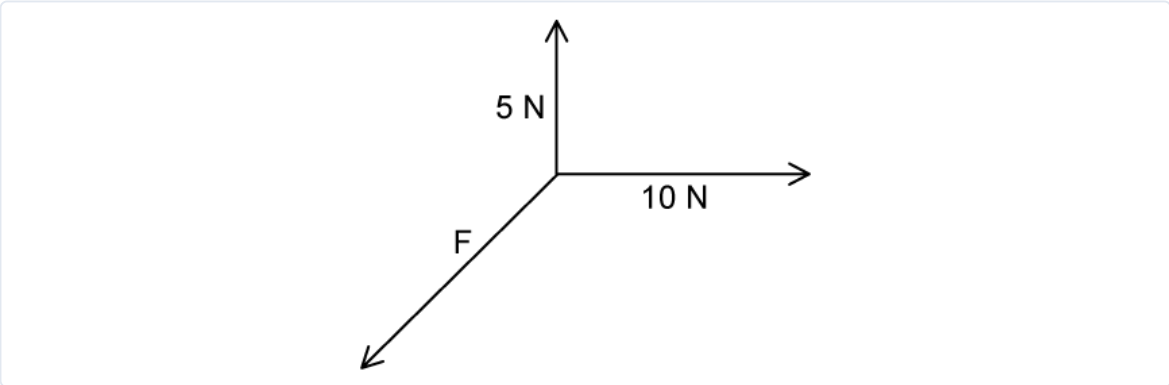
B

C

D

Question 10 Medium [1 mark]

Fig 1.1 shows three coplanar forces which are in equilibrium.



What is the value of force *F*?

A 5 N

B 8.7 N

C 11 N

D 15 N

Question 11 Medium [1 mark]

A ship's speed in still water is 20 km h^{-1}. A current coming from the east pushes the ship at a constant speed of 5 km h^{-1}. At what bearing should the captain direct the ship so that it travels perfectly due north?

A 014.5°

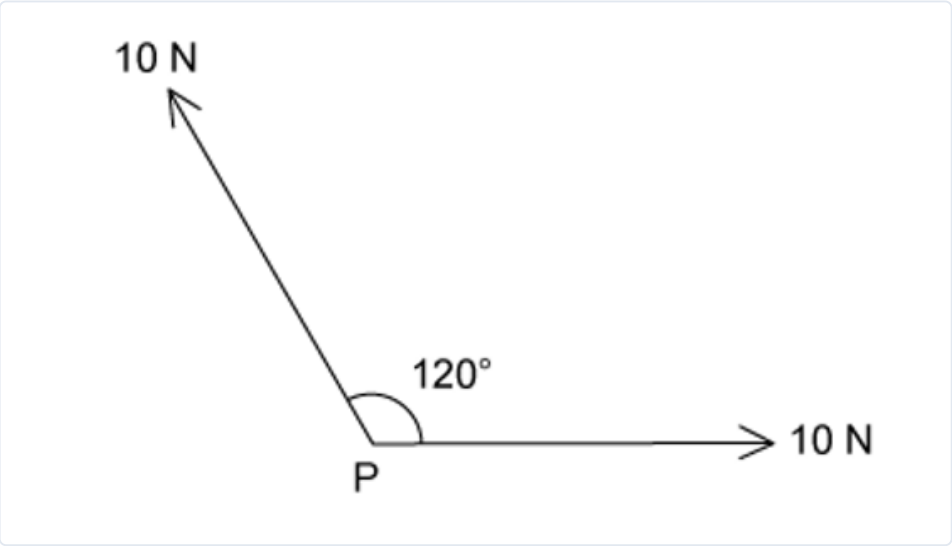
B 075.5°

C 284.5°

D 345.5°

Question 12 Hard [1 mark]

Two forces, each of 10 N, act at a point *P* as shown in the diagram. The angle between the force vectors is 120°



What is the magnitude of the resultant force?

A 5 N

B 10 N

C 17 N

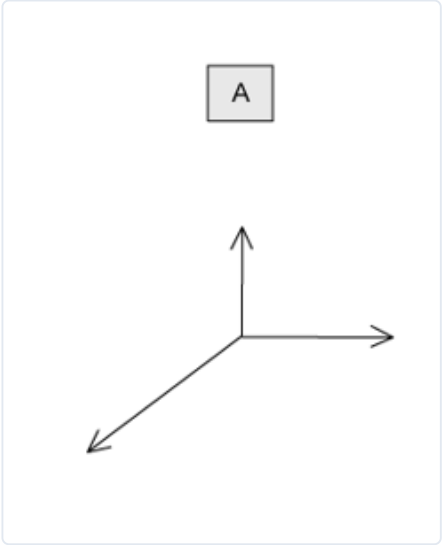
D 20 N

Question 13 Hard [1 mark]

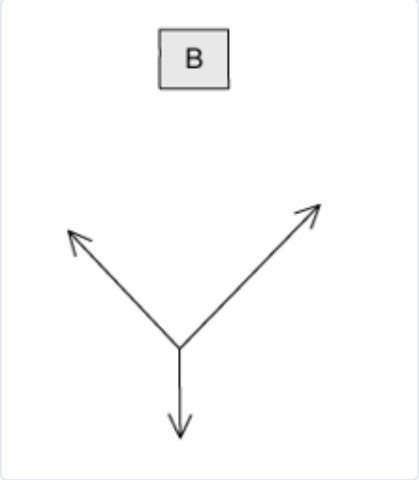
The diagrams represent systems of coplanar forces acting at a point. The lengths of the force vectors represent the magnitudes of the forces.

Which system of forces is in equilibrium?

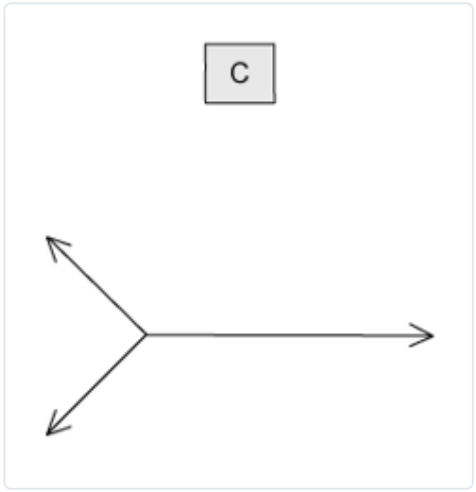
A



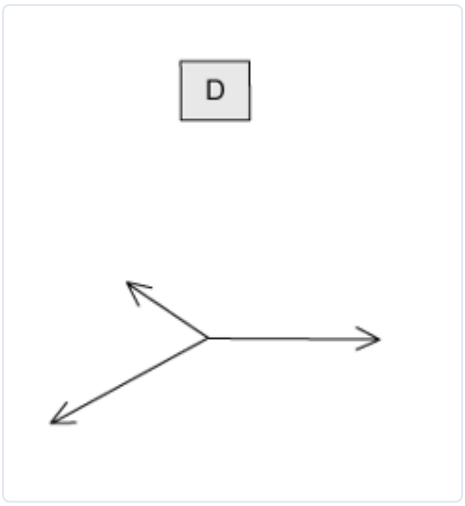
B



C

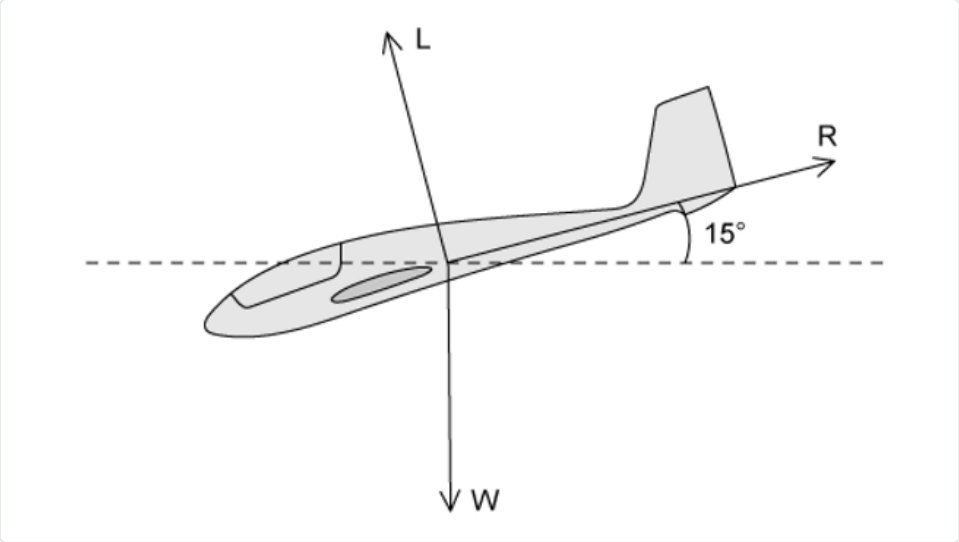


D



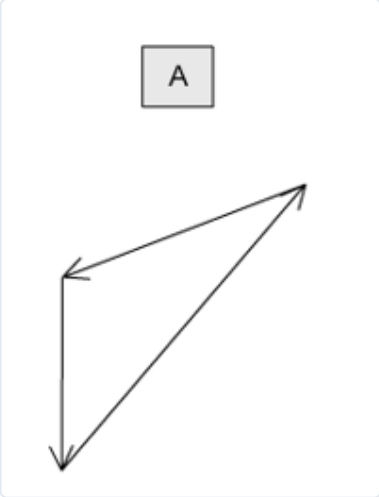
Question 14 Hard [1 mark]

A glider is descending at a constant speed at an angle of 15° to the horizontal. The diagram shows the directions of the lift **L**, air resistance **R** and weight **W** acting on the glider.

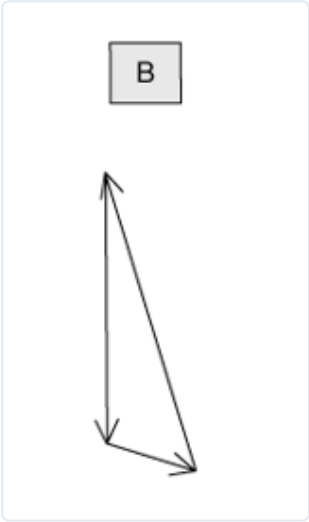


Which vector triangle could represent the forces acting on the glider?

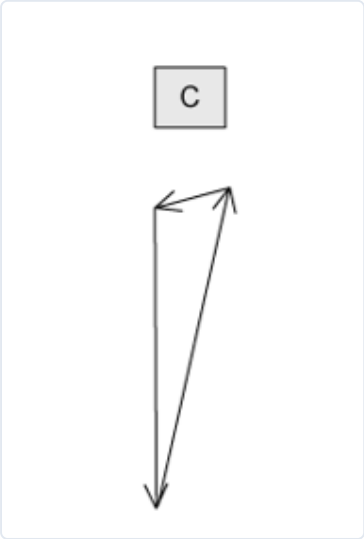
A



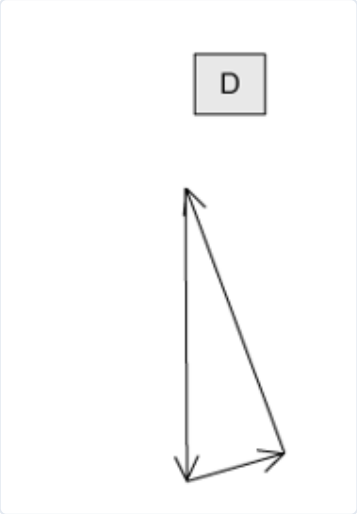
B



C

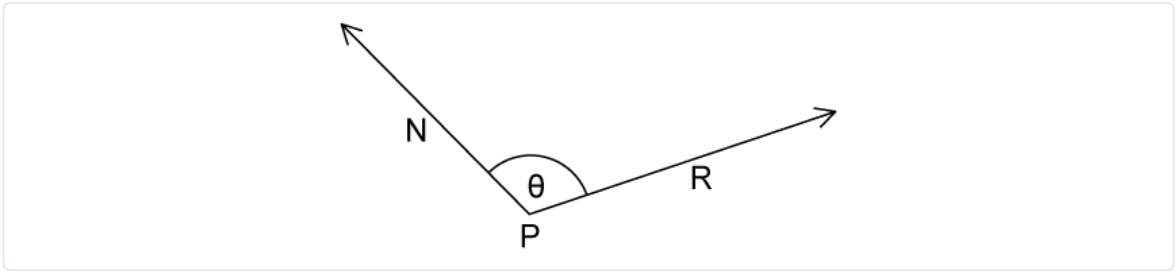


D



Question 15 Hard [1 mark]

Figure 1.1 shows two coplanar forces, *R* and *N*, which act at point *P*. The angle between the forces is θ, where $\theta > 90$.



What is the magnitude of the resultant force *F* equal to?

A $\sqrt{R^2 + N^2 - F^2}$

B $\sqrt{R^2 + N^2 - 2RN \cos (180 - \theta)}$

C $\sqrt{R^2 + N^2 - 2RN \cos (\theta)}$

D $R^2 + N^2 - 2RN \sin(F)$

Question 16 Hard [1 mark]

A boat is being pulled along by two equal forces of 700 kN, both at angle θ to the direction of motion. The boat experiences a resistive force of 40 kN. The net force on the boat is 1270 kN.

What is the value of θ?

A 21°

B 25°

C 57°

D 61°